

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSEAI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Architectural Modelling (CO2)

Group No : 6

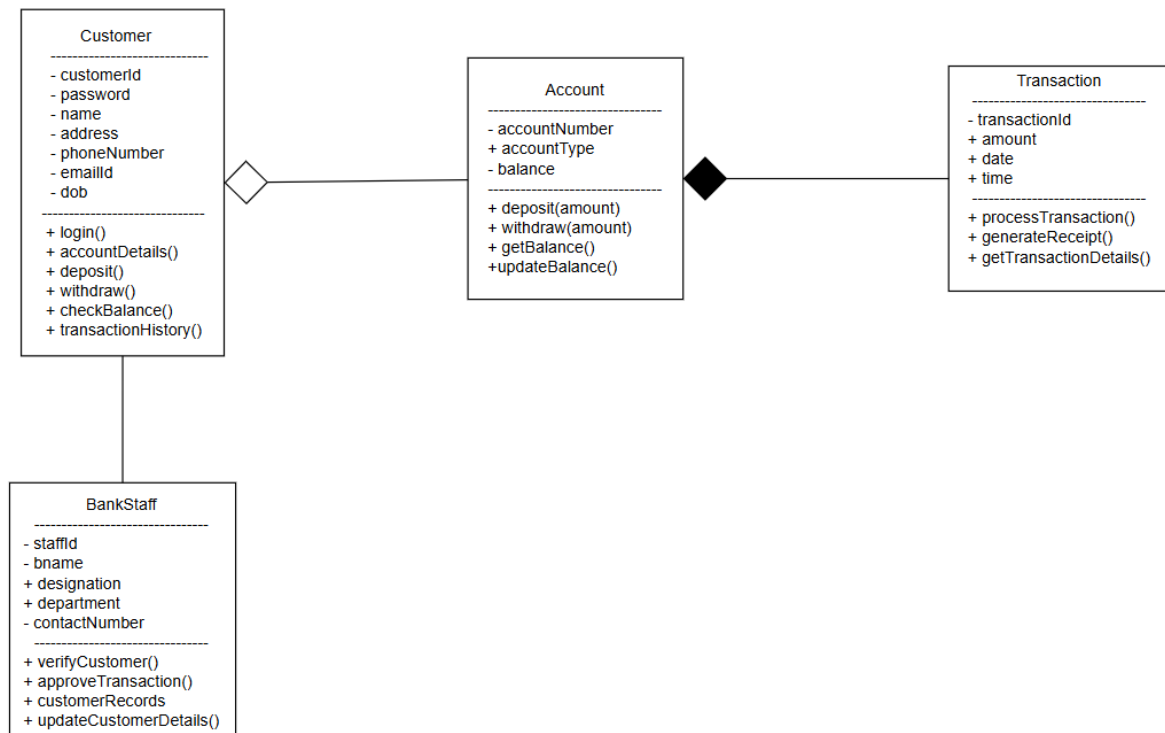
Team Members :

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1. Problem Title: Banking Account Management System

STRUCTURAL DESIGN

2. Class diagram



BEHAVIOURAL DESIGN

3. Use-case diagram

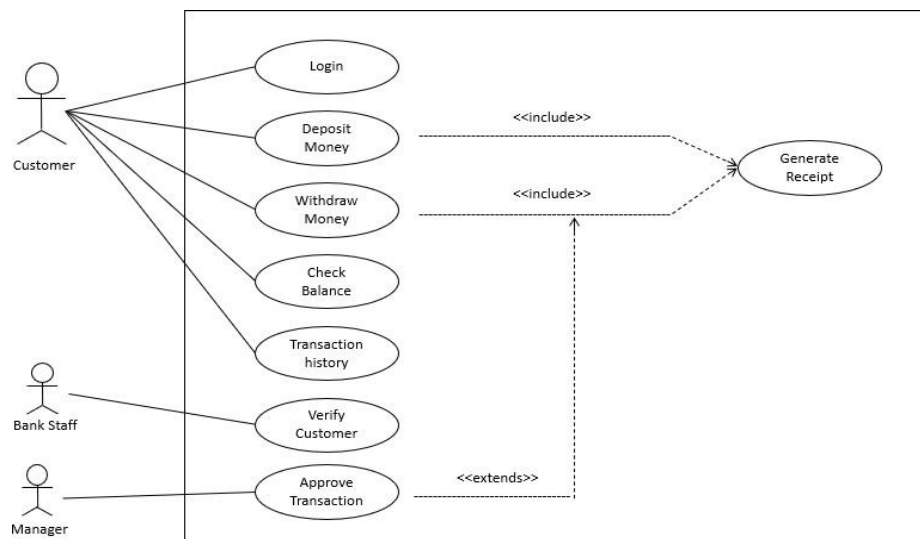
a. List of actors

1. Customer
2. Bank Staff
3. Manager
4. System

b. List of use-cases

1. **Login** – Customer logs into account securely.
2. **Deposit Money** – Customer adds money to account.
3. **Withdraw Money** – Customer withdraws money.
4. **Check Balance** – Customer views account balance.
5. **View Transaction History** – Customer checks previous transactions.
6. **Verify Customer** – Staff verifies customer details.
7. **Approve Transaction** – Staff approves large/special transactions.
8. **Generate Receipt** – System generates receipt after transaction.

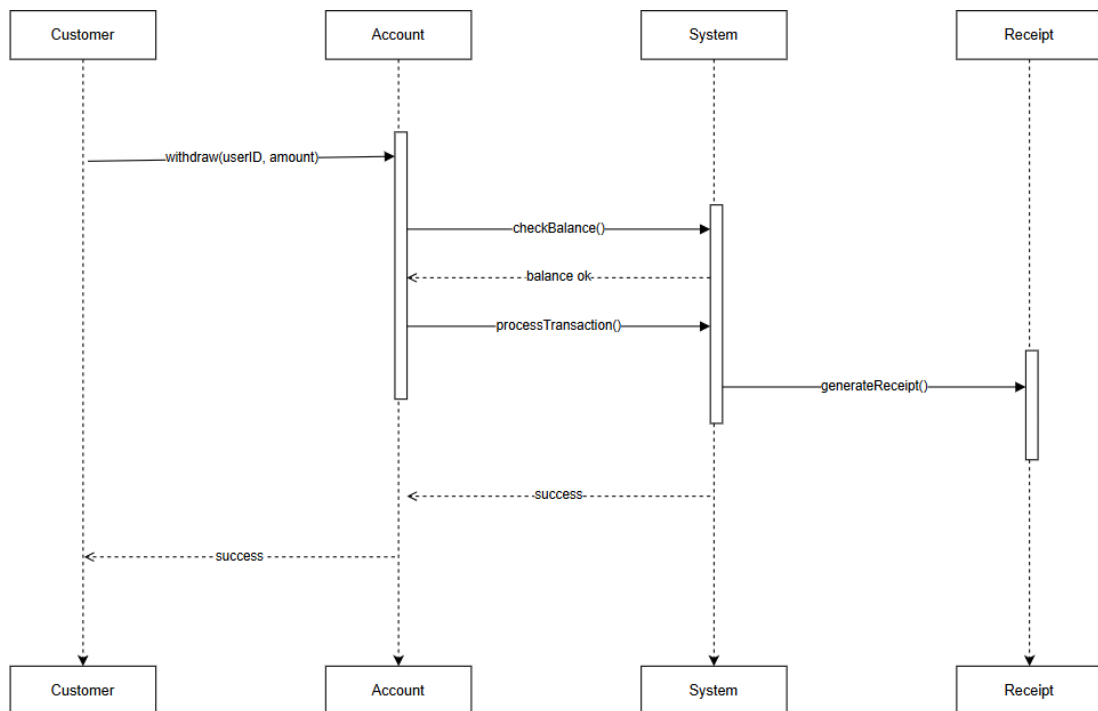
c. Use case diagram



4. Sequence diagram

Withdraw money:

A customer logs into the banking system and requests to withdraw money from the account. The system checks the available balance. If sufficient balance exists, the amount is deducted and the transaction is processed. Finally, the system generates a receipt and shows confirmation to the customer.



Deposit Money:

A customer wants to deposit money into their bank account. The customer enters the account details and deposit amount. The system verifies that the account is valid, updates the account balance, and records the transaction. After the deposit is completed successfully, a receipt is generated and a confirmation message is shown to the customer.

